

The external forcings: air temperature and solar radiation

Characterisation of ERA5 and the Gubbio meteorological station, and their compatibility with the on-structure measurements

Study report · studies/proxy_forcing_characterization/

Abstract

Air temperature and solar radiation are candidate environmental drivers of the monitored wall response, but their on-structure records do not cover the full monitoring period. This report characterises two longer external sources—ERA5 reanalysis retrieved through Oikolab and observations from the Meteosystem station in Gubbio—before comparing them with the final valid on-structure values produced by Study 01. The comparison is restricted to air temperature and solar radiation and asks whether either external source is sufficiently compatible with the local measurements to substitute for them where the local record is absent. Compatibility is assessed without treating the three sources as physically equivalent measurements.

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1 Introduction

The structural interpretation developed after Study 01 requires environmental series that extend across the monitoring period. The measurements made at the wall cannot provide that continuity. The on-structure air-temperature channel contains substantial gaps, while solar radiation was introduced only with the current instrument package and includes periods rejected by the quality

tests in Study 01. External data are therefore considered as possible substitutes, but their greater coverage does not by itself establish that they represent the thermal forcing at the wall.

This report considers two external sources. ERA5 is a gridded atmospheric reanalysis and represents an area rather than the monitored surface. The Meteosystem record is produced by a meteorological station in Gubbio and is a point observation, but it is not made at the wall and its exposure differs from that of the on-structure instruments. Each source is first characterised on its own terms, including its provenance, variables, coverage, complete time series, and summer and winter mean daily cycles. A later comparison then brings ERA5, the meteorological station, and the on-structure measurements onto a common hourly grid.

Source	Nature	Native interval	Documented extent	Role in this study
ERA5 through Oikolab	Gridded atmospheric reanalysis, representing a grid-cell average around the monitoring coordinates	1 hour	1 Jan 2018–12 Aug 2026	Long-coverage proxy for air temperature and solar radiation
Meteosystem Gubbio	Ground-based meteorological station in Gubbio; a point measurement away from the monitored wall	30 minutes	17 Oct 2018–12 Aug 2026	Local meteorological proxy for air temperature and solar radiation

Table 1: The two external sources. The extents are those recorded in the current repository data dictionary and will be recomputed from the input files when the study is executed. Coverage percentages are reported separately for each variable because a source may exist while an individual channel is missing.

2 ERA5 reanalysis

2.1 Nature and coverage

ERA5 combines numerical weather prediction with assimilated observations to produce a spatially continuous reconstruction of atmospheric conditions. The series used here was extracted at the monitoring coordinates (43.353430° N, 12.582047° E) and is described in the repository as representing a grid cell of approximately nine kilometres. It is hourly and currently spans 1 January 2018 to 12 August 2026. The repository data dictionary reports approximately 99.5 % coverage for the ERA5 variables used by this study; the executed analysis will replace that provisional figure with counts measured directly from the input file.

The gridded value is not expected to reproduce the microclimate of a sun-exposed instrument enclosure or masonry surface. Its possible value is instead continuity: if its temperature and radiation retain a stable relation with the on-structure measurements over their shared window, ERA5 may provide a consistent forcing proxy outside that window.

2.2 Retrieval and repository organisation

The paired notebook and script `auxiliary/oiko.ipynb` and `auxiliary/oiko.py` retrieve the data from the authenticated Oikolab weather API. The script requests hourly means in six-month chunks, concatenates the responses, removes duplicate timestamps and provider metadata columns, and appends only dates later than the current local export. The consolidated file is

written to `data/raw/proxies/oikolab_weather.csv`; Study 02 reads that file without making a new network request. Timestamps are supplied in UTC.

2.3 Variables

Table 2 records the variables requested by the retrieval script that are retained for this report and their interpretation. Every variable in the table is included in the source characterisation. Only air temperature and surface solar radiation subsequently enter the three-source comparison.

Oikolab variable	Unit	Meaning
temperature	°C	Near-surface air temperature; compared with the on-structure and Meteosystem air-temperature series.
relative_humidity	fraction	Relative humidity. It is multiplied by 100 on loading when expressed as a fraction.
dewpoint_temperature	°C	Temperature at which the represented air mass reaches saturation.
surface_solar_radiation	W/m ²	Downward short-wave solar radiation at the surface; compared with the other radiation series.
total_precipitation	mm water equivalent	Accumulated liquid and solid precipitation expressed as water depth.
wind_speed	m/s	Magnitude of the near-surface wind.
wind_direction	[angular convention required]	Direction from which the wind originates. It will be normalised to degrees and treated as a circular variable.

Table 2: ERA5 variables retained for characterisation. The unit, zero direction, and rotation convention of wind direction must be confirmed from the API response metadata before the table is finalised.

2.4 Complete time series

The complete ERA5 record is shown before any restriction to the on-structure measurement window. The seven retained variables occupy separate vertically aligned panels with a shared time axis, each panel named by its own title and carrying its unit on the axis. They are ordered air temperature, relative humidity, solar radiation, wind speed, wind direction, dew-point temperature and total precipitation: the first five are the channels the daily-cycle figures draw, in the same order, so that a reader moving between the figures of this report finds each channel where it was left. Gaps remain visible rather than being connected across, and wind direction is drawn on a circular angular scale. This view establishes the actual extent, missing intervals, long-term range, and seasonal envelope of every retained variable.

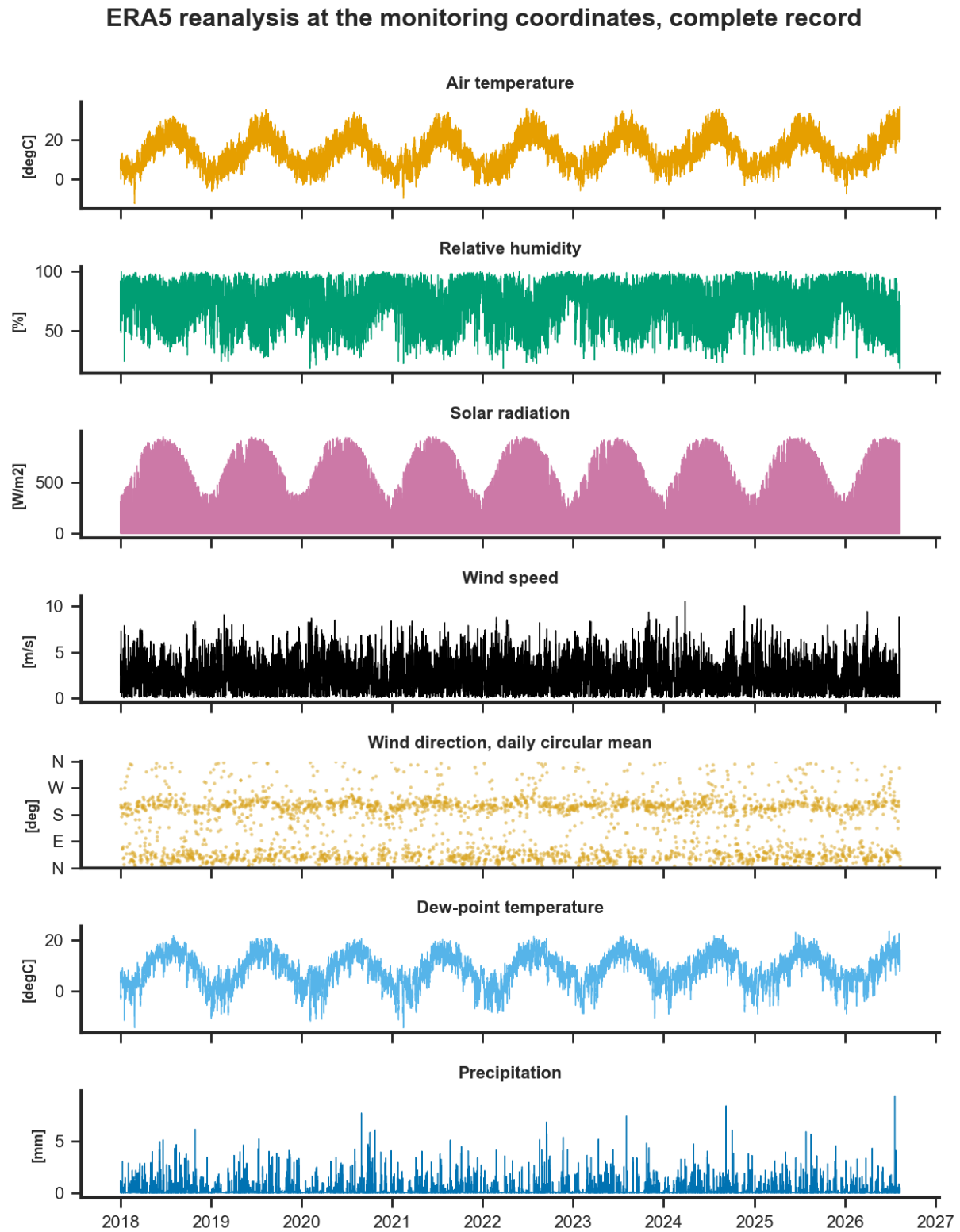


Figure 1: Complete ERA5 record, one panel for each of the seven retained variables on a shared time axis. Wind direction is drawn as its daily circular mean, since eight years of hourly bearings fill a panel solid.

2.5 Summer and winter mean daily cycles

Mean daily cycles are computed separately for summer and winter from the complete valid ERA5 record. Summer is provisionally June–August and winter is provisionally December–February, matching Study 01. Each mean is supported by its contributing day count so that a poorly covered

hour is not presented as equally certain to a well-covered one. Scalar variables use ordinary hourly means. Wind direction uses a circular hourly mean after its angles have been normalised to one documented convention.

The two seasons are drawn in one figure rather than two, a channel to a row and a season to a column, and the two panels of a row share one vertical scale. The seasonal difference in a channel's day is then a difference in the drawn shape, rather than one a reader must reconstruct from two axes with different limits on two separate pages. Five of the seven retained channels are drawn: the two forcings this study is about, the humidity that accompanies them, and the wind that ventilates the surface. Dew point and precipitation keep their place in the complete-record figure of Section 2 and in the tabulated diurnal profile, both of which cover every retained channel; what is narrowed here is the figure, not the characterisation.

ERA5, mean daily cycle by season (June-August, December-February)

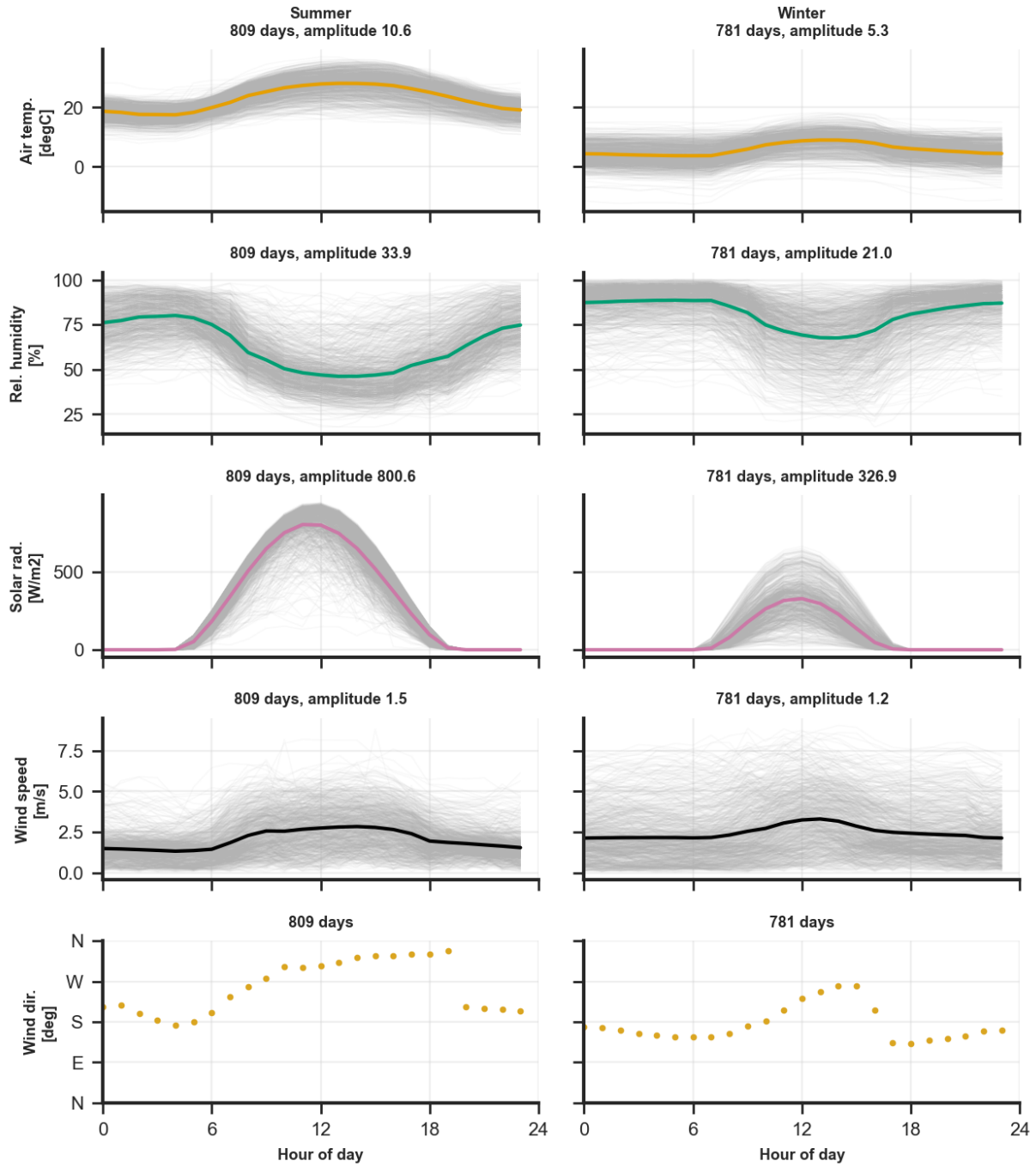


Figure 2: ERA5 mean daily cycles, by channel and season. Air temperature, relative humidity, solar radiation, wind speed and wind direction, one to a row, with summer in the left column and winter in the right. Every contributing day is drawn in gray behind its mean, and each panel names the days its mean rests on and the peak-to-trough amplitude of that mean.

3 The Gubbio meteorological station

3.1 Nature and coverage

The Meteosystem source is a ground-based station in Gubbio. In contrast to the spatial average represented by ERA5, it records conditions at one location. That location is closer to the monitored

structure than a regional grid is, but distance alone does not establish representativeness: elevation, surrounding surfaces, shading, sensor height, and screen exposure may all separate the station record from the microclimate at the wall.

The downloaded record is natively half-hourly and currently spans 17 October 2018 to 12 August 2026. The repository data dictionary reports 88.6 % coverage for air temperature and 88.7 % for solar radiation. These values will be recomputed by the study because the source is updated month by month and the two variables need not share the same missing intervals.

Information required

Insert the station’s coordinates, elevation, distance and height difference from the monitored wall, sensor models, meteorological exposure, calibration information, operator, and maintenance or relocation history.

3.2 Retrieval and repository organisation

The paired notebook and script `auxiliary/meteosystem_italy.ipynb` and `auxiliary/meteosystem_italy.py` retrieve one CSV file per calendar month from the Meteosystem website. Each response is retained in `auxiliary/data/raw/proxies/_cache/gubbio/`, allowing later cleaning and assembly without another download. The monthly files are concatenated, deduplicated, and converted from Italian civil time to UTC. The current preparation script removes all categorical direction fields before writing `data/raw/proxies/meteosystem_gubbio.csv`. For this characterisation it will be revised to retain `Dir`, convert its compass sectors to a documented angular convention, and remove only the unselected fields. Study 02 aggregates scalar half-hourly observations to hourly means and computes wind direction by a circular hourly mean.

3.3 Variables

Table 3 describes the fields supplied by the station. Every field in the table is included in the station characterisation. Air temperature and solar radiation alone enter the later compatibility study with ERA5 and the on-structure measurements.

Raw field	Unit	Meaning
Temp	°C	Air temperature; compared with ERA5 and the on-structure air-temperature series.
Umid	%	Relative humidity.
Dew pt	°C	Dew-point temperature.
Vento	m/s	Mean wind speed.
Dir	compass sector	Mean wind direction. Compass sectors will be converted to angles and treated as circular data.
Press	hPa	Atmospheric pressure.
Pioggia	mm	Precipitation depth.
Int.Pio.	[unit required]	Precipitation intensity.
Rad.Sol.	W/m ²	Solar radiation; compared with ERA5 and the on-structure radiation series.

Table 3: Variables in the Meteosystem monthly files. The reporting interval and unit of precipitation intensity require confirmation from the station documentation.

3.4 Complete time series

The complete station record is shown independently of the shorter on-structure record, on the layout and the channel order used for ERA5 in Figure 1: air temperature, relative humidity, solar

radiation, wind speed, wind direction, dew-point temperature, precipitation depth and atmospheric pressure, each panel named by its own title and carrying its unit on the axis. The plot preserves missing intervals and uses the half-hourly observations after the quality-control decisions that produce the consolidated source file. Wind direction is shown on a circular angular scale.

Precipitation intensity is the ninth retained channel and is not drawn. It is the precipitation depth of the panel beside it divided by the reporting interval, so the two panels would carry one measurement at two scales. It remains in the channel inventory and in the tabulated diurnal profile, both of which cover all nine.

The precipitation panel is drawn on a clipped vertical axis. The station's precipitation depth contains a single reading of 1.70×10^9 mm in one hour, and its intensity channel contains five readings of the same order; the largest depth that is physically possible is the next value down, at 30.1 mm. An axis scaled to the defective sample renders the entire eight-year record as a flat line beneath one spike, so the axis is bounded by the values inside the channel's documented plausible range and the panel carries a note, in the accent colour, stating how many samples fall outside it and how far. The clipping is a display choice: no value is removed from the data, and the implausible readings are counted and reported in the channel inventory (Table 3 describes the fields; the executed inventory records one implausible depth and five implausible intensities).

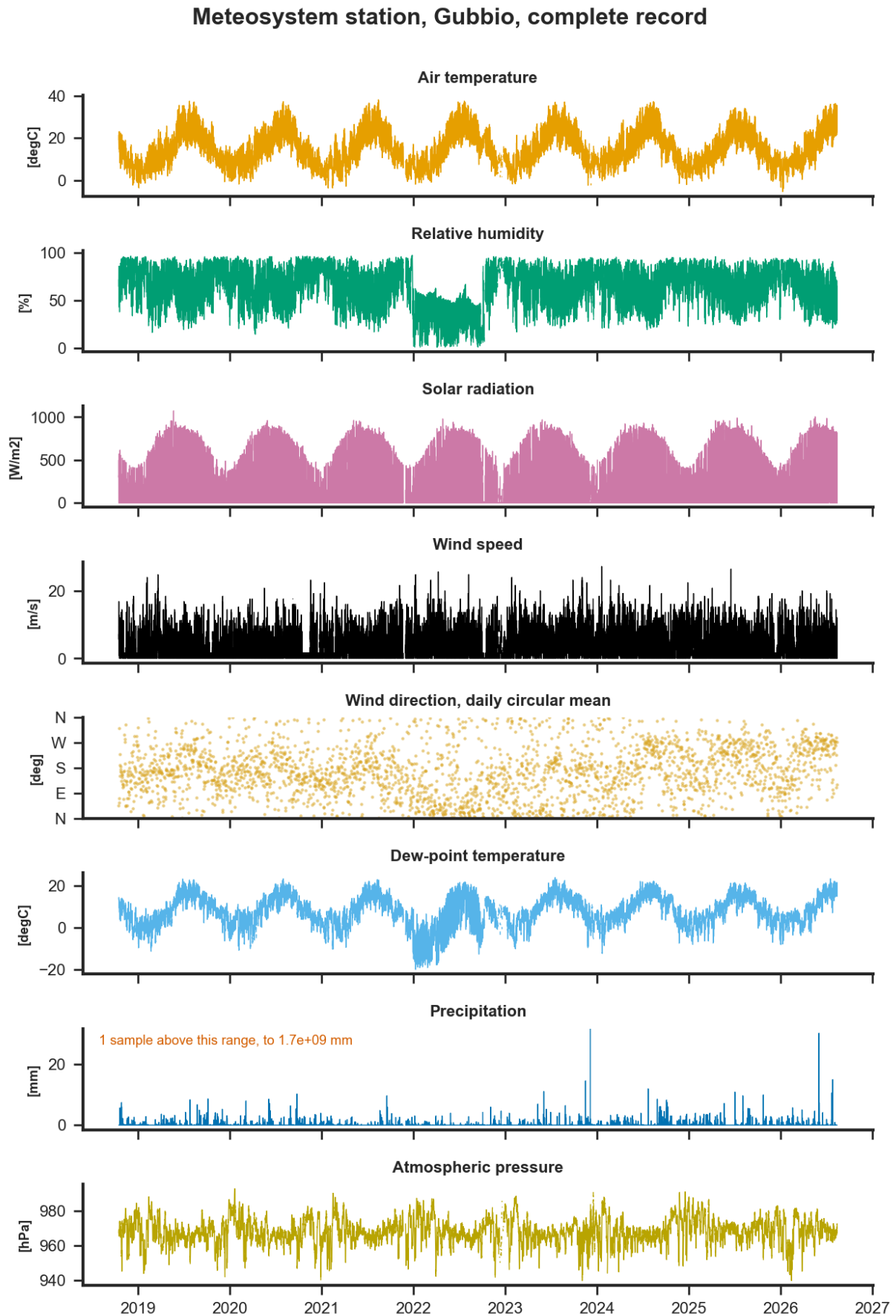


Figure 3: Complete Meteosystem record, one panel for each of eight retained variables on a shared time axis, in the channel order used for ERA5. Precipitation intensity is omitted as a rescaling of the precipitation depth above it. The precipitation axis is clipped to the channel’s plausible range, with a note giving the sample it excludes. Wind direction is drawn as its daily circular mean.

3.5 Summer and winter mean daily cycles

The seasonal profiles are computed from the complete valid station record, before restricting it to timestamps shared with the on-structure instruments. The season definitions and display conventions match those used for ERA5, so differences between the two characterisations reflect the records rather than different summaries. Scalar variables use ordinary hourly means. Wind direction uses a circular hourly mean of the converted angles.

The layout is the one used for ERA5 in Figure 2 and draws the same five channels in the same order, so that the two sources can be read against each other panel by panel. The station’s four remaining channels—dew-point temperature, atmospheric pressure, precipitation depth and precipitation intensity—keep their place in the complete-record figure of this section and in the tabulated diurnal profile, both of which cover all nine.

Meteosystem station, mean daily cycle by season (June-August, December-February)

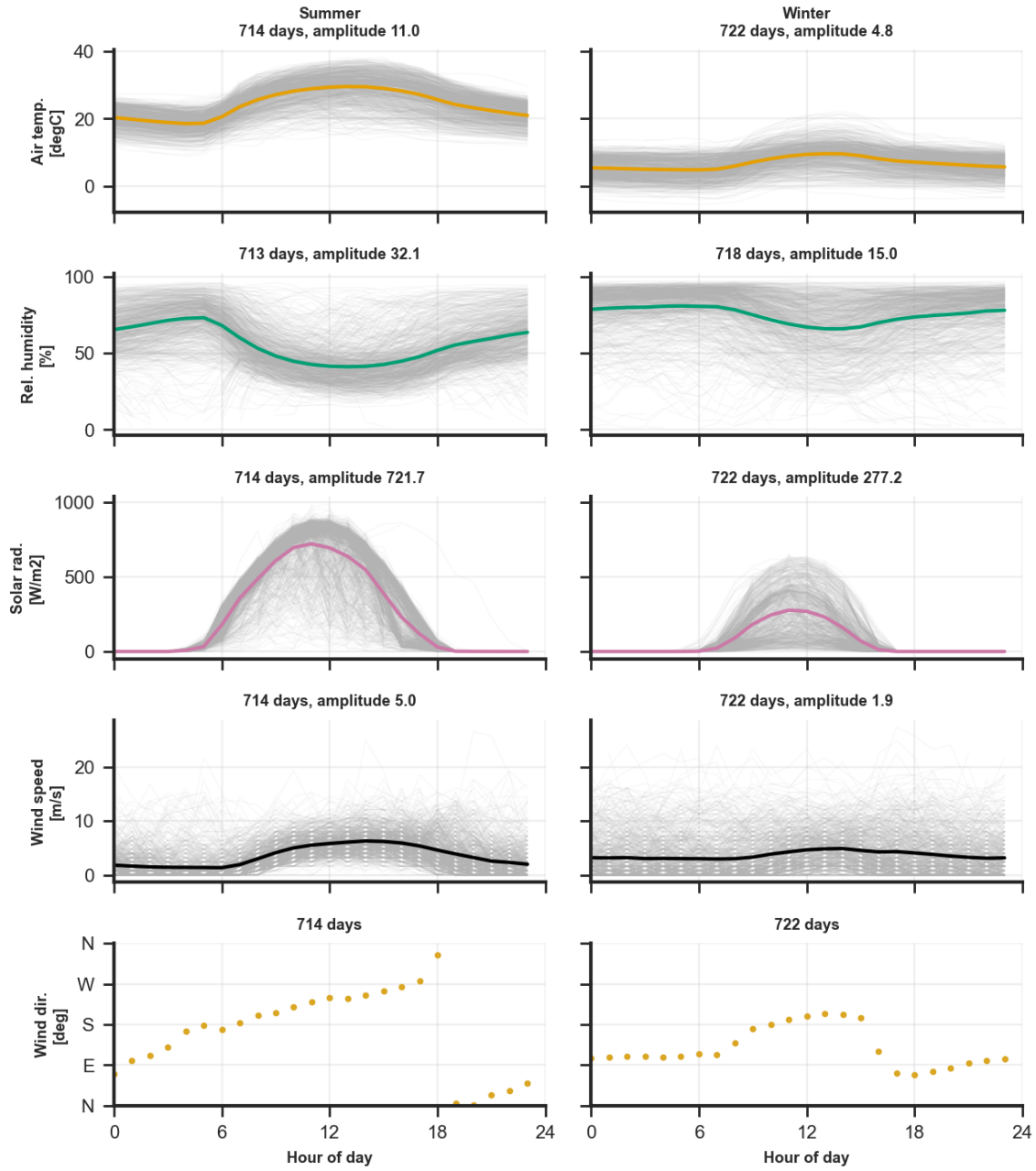


Figure 4: Meteosystem mean daily cycles, by channel and season. Air temperature, relative humidity, solar radiation, wind speed and wind direction, one to a row, with summer in the left column and winter in the right, on the layout and the channel order used for ERA5. Every contributing day is drawn in gray behind its mean, and each panel names the days its mean rests on and the peak-to-trough amplitude of that mean.

4 The two external sources against each other

Both external sources are now on the common hourly grid, and each has been characterised on its own terms. Before either is measured against the wall, they are measured against each other, over every quantity they both report.

This comparison answers a question the on-structure comparison cannot. The wall record is short, and for radiation it is trustworthy over a narrower window still, so a disagreement found

there says little about how the two sources behave outside it. Against each other they have their entire shared record and every quantity in it, not only the two forcings. What that establishes is the kind of source each one is: where a grid-cell average of roughly nine kilometres departs from a point observation, in which direction, and whether the departure is the same in summer as in winter. A quantity on which the two agree closely is one where the later choice between them will not matter much; a quantity on which they diverge is one where the verdict of Section 5 has to be earned rather than assumed.

Agreement is reported against the station, so that a positive bias reads as the reanalysis running high with respect to the local observation. Neither source is a reference in the sense the on-structure record is: this is a comparison of two proxies, not the validation of one against a truth. Every pair uses its own overlap, and no value is interpolated to manufacture one. Each quantity is reported over the whole shared record and then separately for each season, because a bias that is stable across the year and a bias that changes sign with the season are indistinguishable in a statistic pooled over both.

The scores are computed on values inside each quantity's documented plausible range. This is the one place in the study where the implausible readings of Section 3 are set aside rather than merely counted, and the reason is that a score cannot survive them: a single rainfall reading of order a billion millimetres dominates every mean square it enters, and the resulting row would look like a result while being an artefact of one defective sample. Nothing is deleted. The readings remain in the frame every other step of the study uses, remain counted in the channel inventory, and remain visible in the complete-record figure; what is excluded here is their effect on a correlation and a root mean square, and the number of values excluded is reported with the table.

Precipitation carries one further exclusion, and it is of a different kind. The station's heaviest recorded hour, 30.1 mm, is real weather: it is well inside the plausible range, roughly twice the next largest hour in eight years, and is the sole occupant of the upper two thirds of its panel's axis. For this comparison alone the precipitation channels are bounded at 20 mm in the hour, which excludes that single observation, so that the row and the panel describe the behaviour the two sources ordinarily produce rather than one storm. This is a decision about what the comparison covers, not the removal of a defect, and it is stated here because a reader who assumed otherwise would misread the precipitation row. The hour is retained everywhere else in this report, and the bound is a parameter of the study: widening it puts the observation back.

Wind direction is differenced on the circle, as everywhere else in this study. Its bias and correlation are the columns to read on that row: the amplitude ratio and the phase lag are computed by the same shared machinery as for the scalar quantities, where for a bearing they do not carry a meaningful peak-to-trough distance, and the ratio is reported as missing. It is missing for precipitation too, whose daily cycle has no amplitude to take a ratio of on a day without rain.

Three figures accompany the table, each answering a question the others average away. The first is the marginal view: every hour on which both sources report, one quantity to a panel, against the 1:1 line. A constant offset appears as a cloud displaced parallel to the diagonal, and a source that compresses the range of a quantity appears as a cloud flatter than the diagonal — which is what an amplitude ratio below one looks like before it is reduced to a number.

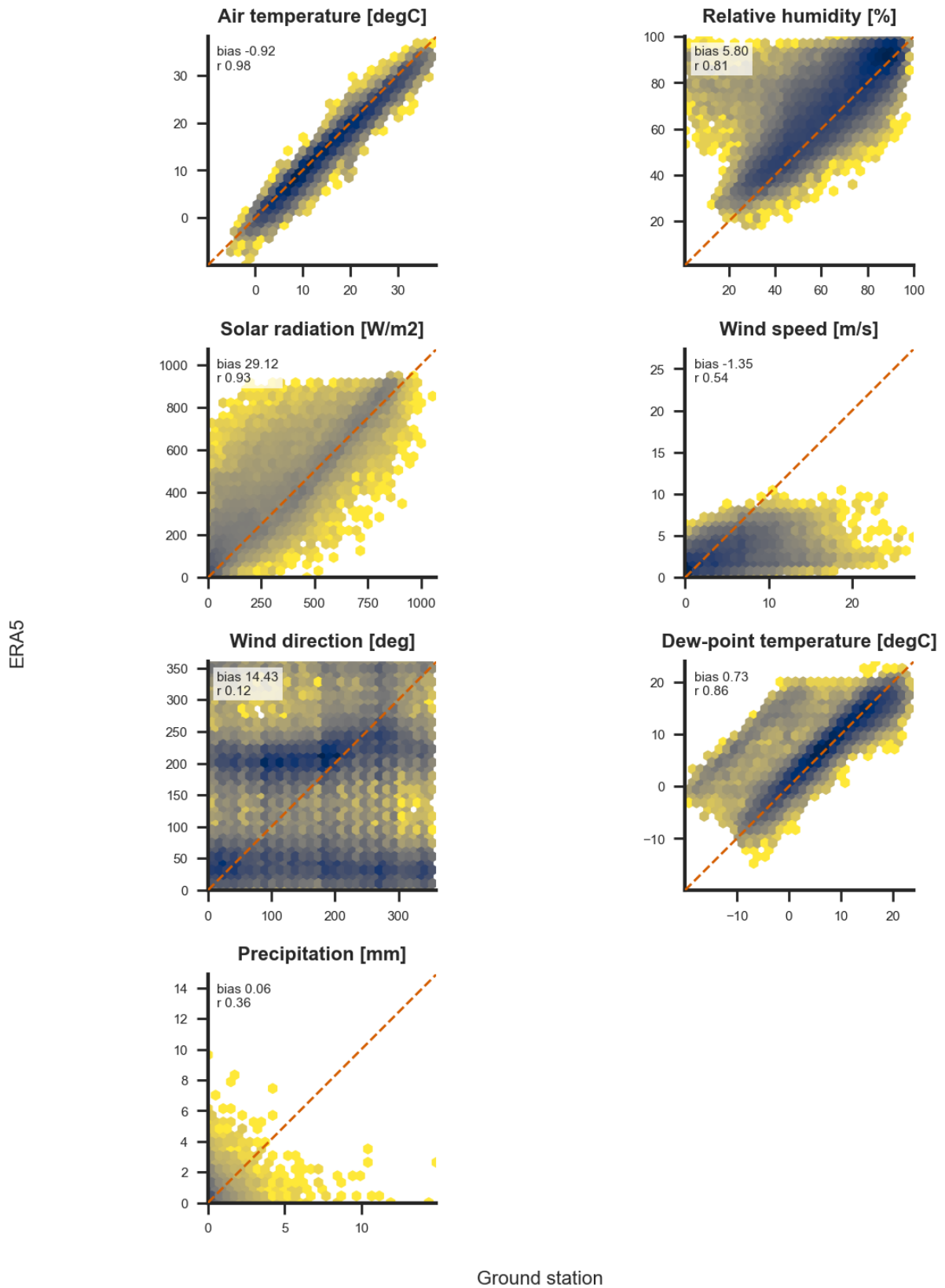


Figure 5: ERA5 against the Meteosystem station, every paired hour, one panel per shared quantity. The dashed line is 1:1 and the bias and correlation printed in each panel are the whole-record values of Table 4, not a second computation. Darker hexagons hold more samples, on a logarithmic count scale. Precipitation is bounded at 20 mm in the hour, excluding the station’s single heaviest observation.

The second asks whether the disagreement holds still. This is the question that decides whether

a source may substitute for another: a constant offset can be subtracted and the substitution documented, while an offset that changes with the season cannot, and the two are indistinguishable in any statistic pooled over the whole record.

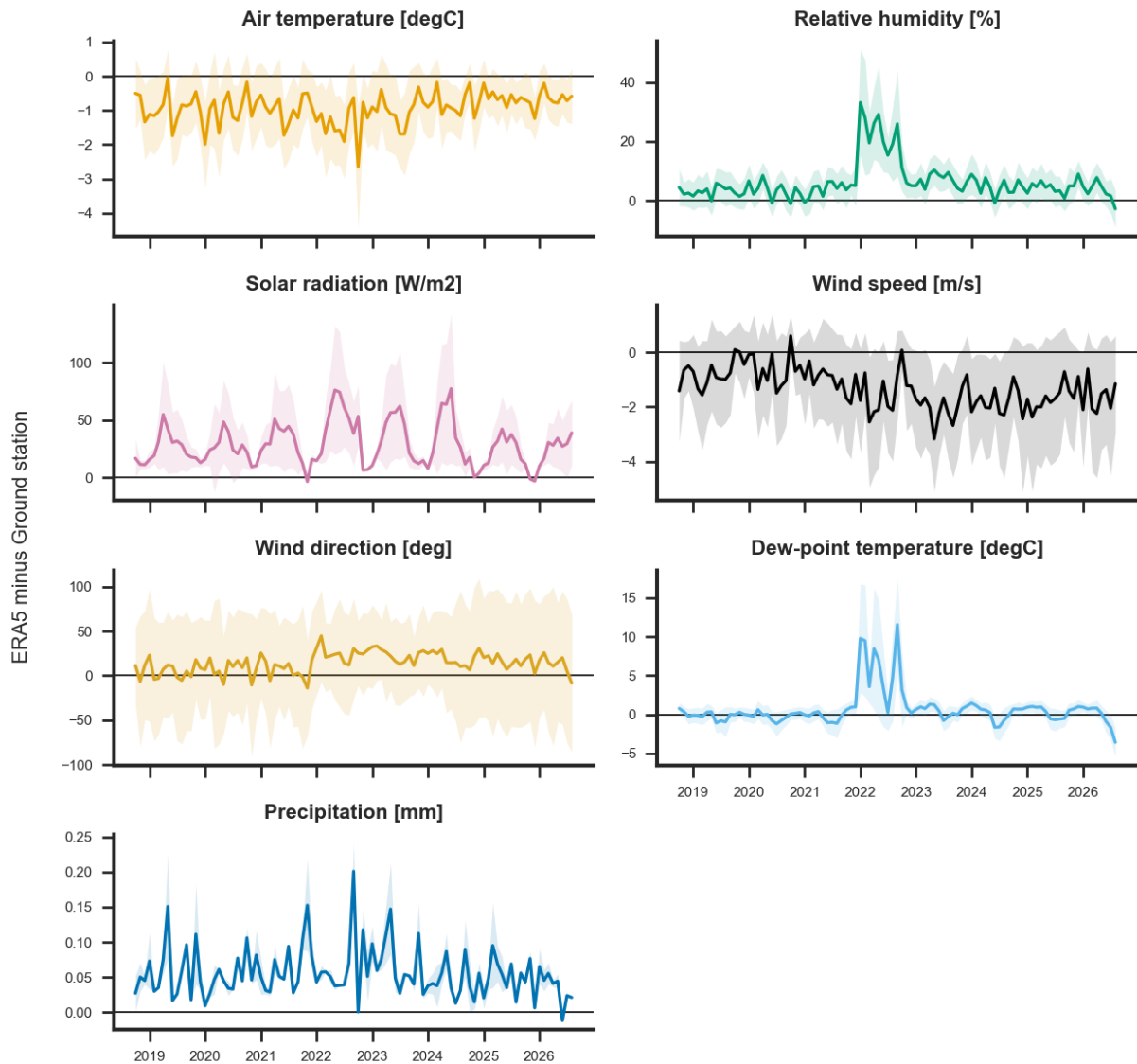
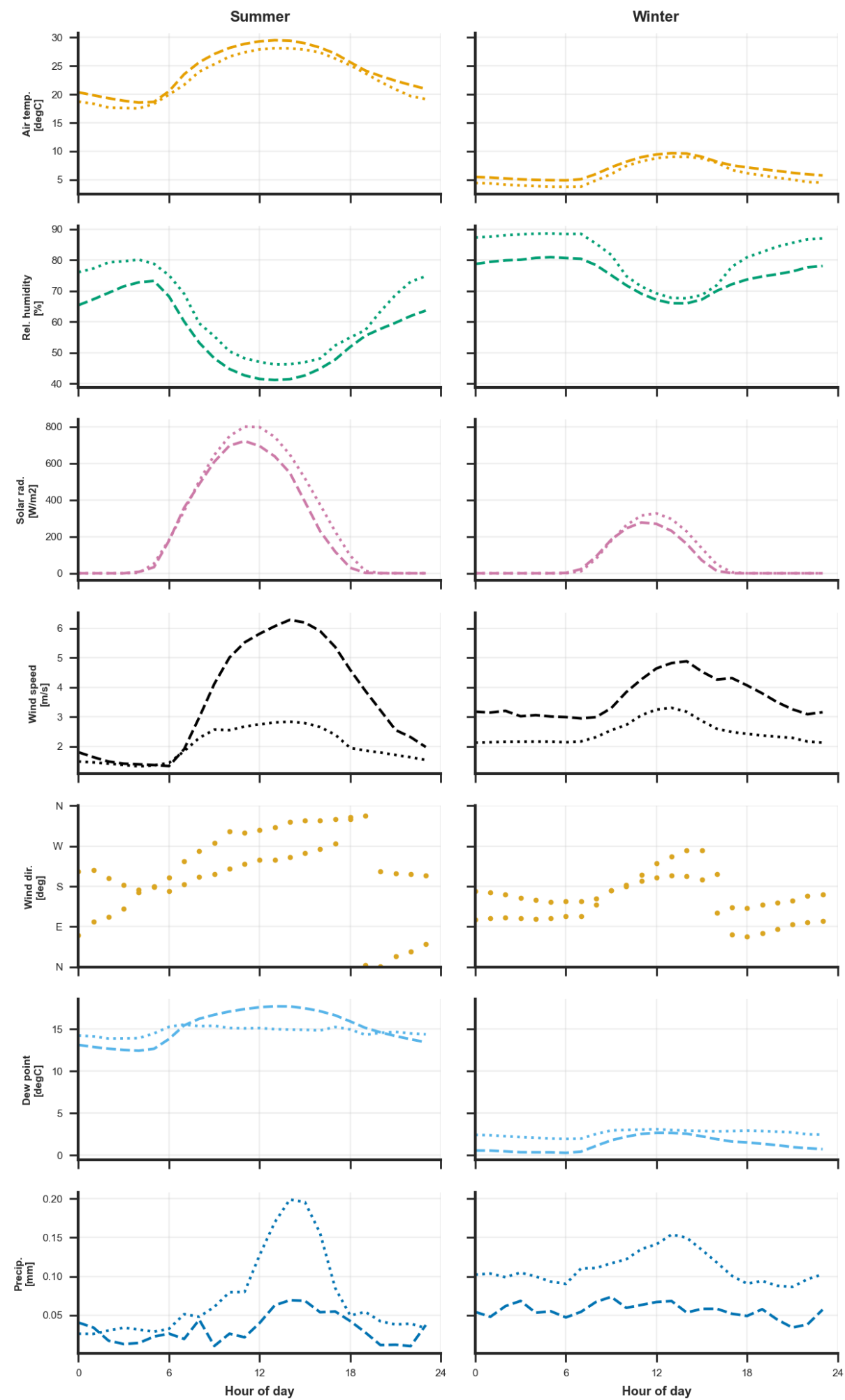


Figure 6: The difference between the two sources, month by month, one panel per shared quantity. The band is the interquartile range of the difference within each month, so a drift and a widening can be told apart. Positive is ERA5 reading high.

The third asks about the day. Two sources may agree on the level of a quantity and still disagree about when it peaks and how far it travels between its extremes, and neither of the first two figures can show that, because both average the time of day away.



Quantity	Split	N	Bias	MAE	RMSE	r	Amp Ratio	Phase Lag
Air temperature	all	66,516	-0.92	1.36	1.81	0.981	1.03	0.44
Air temperature	summer	16,895	-1.14	1.50	1.92	0.958	0.94	0.43
Air temperature	winter	16,293	-0.93	1.34	1.81	0.915	1.12	0.46
Air temperature	spring	17,430	-0.85	1.34	1.80	0.958	1.04	0.46
Air temperature	autumn	15,898	-0.77	1.24	1.69	0.963	1.07	0.42
Relative humidity	all	64,564	5.80	8.80	12.90	0.806	1.11	-1.37
Relative humidity	summer	16,627	6.01	8.93	12.62	0.799	1.01	-1.46
Relative humidity	winter	15,732	5.77	8.96	13.80	0.692	1.22	-1.41
Relative humidity	spring	16,940	6.96	9.64	14.13	0.767	1.09	-1.18
Relative humidity	autumn	15,265	4.31	7.56	10.62	0.811	1.17	-1.36
Solar radiation	all	66,575	29.12	47.36	97.04	0.930	1.02	0.33
Solar radiation	summer	16,895	40.96	58.06	113.04	0.942	1.03	0.24
Solar radiation	winter	16,341	14.45	29.14	62.21	0.894	0.99	0.41
Solar radiation	spring	17,441	43.04	63.89	120.92	0.908	1.03	0.41
Solar radiation	autumn	15,898	16.35	36.59	76.38	0.924	0.98	0.26
Wind speed	all	66,241	-1.35	2.29	3.36	0.538	0.33	-1.21
Wind speed	summer	16,893	-1.49	2.27	3.17	0.500	0.29	-1.47
Wind speed	winter	16,216	-1.17	2.23	3.45	0.579	0.37	-1.45
Wind speed	spring	17,434	-1.66	2.50	3.57	0.513	0.33	-1.23
Wind speed	autumn	15,698	-1.04	2.13	3.24	0.539	0.31	-0.59
Wind direction	all	66,585	14.43	73.59	90.05	0.120	—	-3.03
Wind direction	summer	16,895	11.29	74.02	90.25	0.131	—	-2.63
Wind direction	winter	16,341	20.57	73.58	89.67	0.073	—	-2.51
Wind direction	spring	17,442	14.12	73.20	90.15	0.123	—	-3.60
Wind direction	autumn	15,907	11.79	73.56	90.13	0.121	—	-1.89
Dew-point temperature	all	66,526	0.73	2.04	3.66	0.864	0.81	1.98
Dew-point temperature	summer	16,893	-0.46	2.29	3.46	0.496	0.71	-0.80
Dew-point temperature	winter	16,324	1.28	2.01	3.91	0.706	0.88	4.06
Dew-point temperature	spring	17,417	1.21	2.06	3.76	0.768	0.88	0.92
Dew-point temperature	autumn	15,892	0.90	1.77	3.49	0.730	0.87	3.59
Precipitation	all	66,584	0.06	0.10	0.39	0.359	—	-2.29
Precipitation	summer	16,894	0.04	0.08	0.44	0.227	—	-6.37
Precipitation	winter	16,341	0.05	0.09	0.29	0.553	—	-0.12
Precipitation	spring	17,442	0.07	0.11	0.35	0.362	—	-5.81
Precipitation	autumn	15,907	0.07	0.13	0.48	0.361	—	-0.74

Table 4: ERA5 against the Meteosystem station, over every quantity both sources report, on their shared hourly record. The station is the reference, so a positive bias is the reanalysis reading high. Each quantity appears once over the whole record and once for each season. Units are those of the quantity itself, given in Tables 2 and 3.

5 Comparison with the on-structure measurements

The comparison brings together ERA5, the Meteosystem station, and the final values produced by Study 01. It does not repeat the on-structure data-quality study. Air temperature is read from its final joined and corrected `tair` column. Solar radiation is read from `n_sr_ok`, with all timestamps on days marked by `sr_suspect` excluded. Where Study 01 classified a radiation value as `night`, the corrected value of 0 W/m^2 in `n_sr_ok` is used; the original non-zero sensor floor is not restored.

All three sources will be expressed in common units and placed on an hourly UTC grid. The

twenty-minute on-structure values and the half-hourly Meteosystem values will be aggregated by hourly mean, while ERA5 is already hourly. Every valid on-structure value will remain eligible for comparison. Pairwise statistics will use the overlap with each external source, and direct three-source figures will use the timestamps on which all three are available. No proxy value will be interpolated to manufacture an overlap.

5.1 Air temperature

The temperature comparison will align the three series over every valid on-structure observation. It will show their absolute levels, common temporal variation, and differences. Agreement will be summarised with bias, error, correlation, and differences in daily amplitude and phase, reported by season where the coverage supports that division.

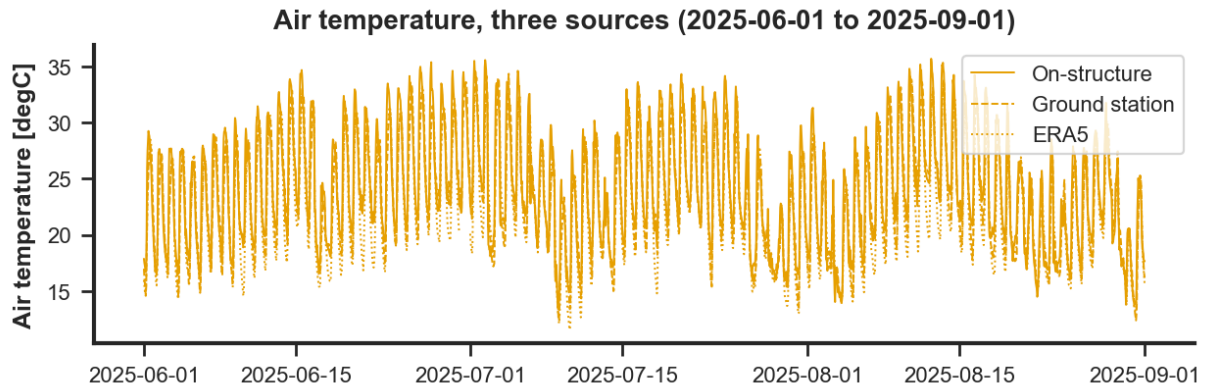


Figure 8: Aligned three-source air-temperature time series over the valid on-structure timestamps.

Air temperature: each external source against the on-structure housing

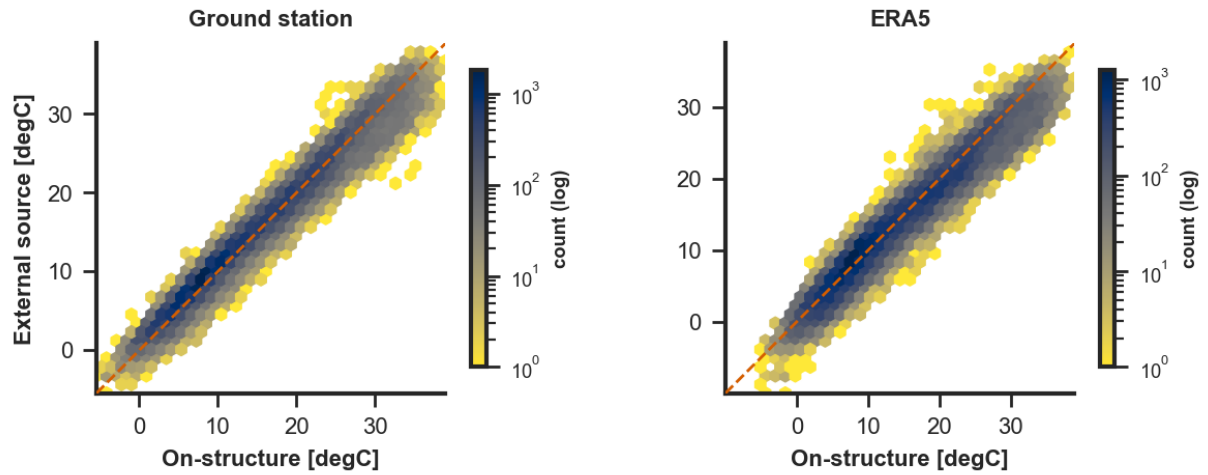


Figure 9: Direct comparison scatter plots of each air-temperature proxy against the on-structure values.

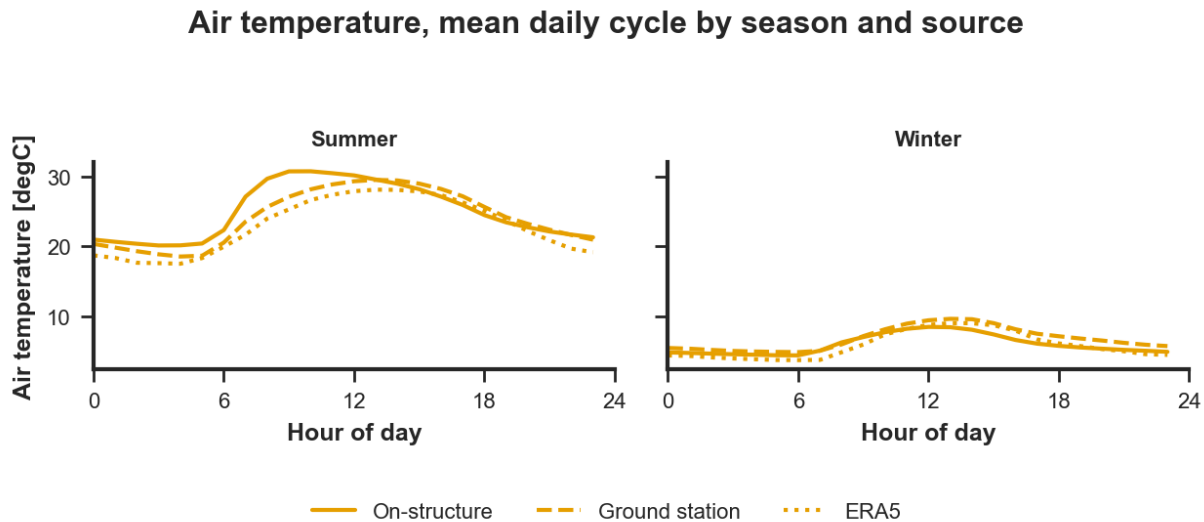


Figure 10: Air-temperature diurnal cycles across sources.

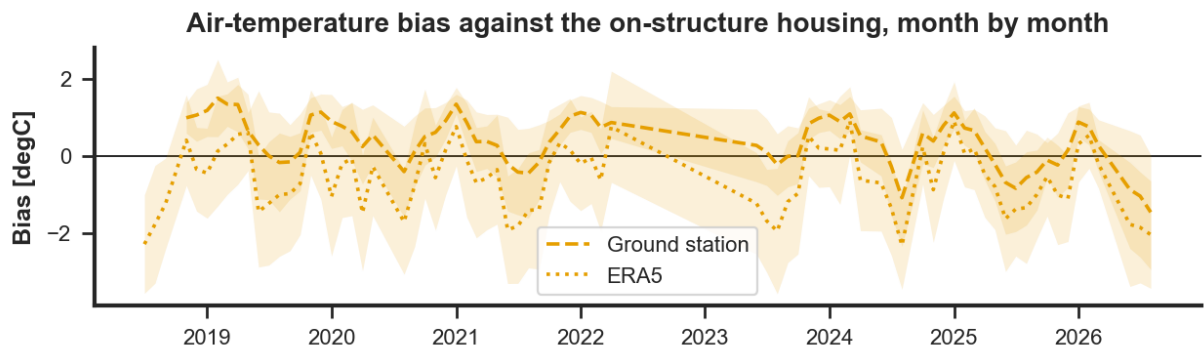


Figure 11: Air-temperature stability and difference analysis.

Source	Season	N	Bias	MAE	RMSE	r	Amp (On)	Amp (Prx)	Amp Ratio	Phase D
gs	all	47,044	0.37	1.30	1.64	0.983	8.02	9.00	1.12	1
era5	all	49,181	-0.56	1.70	2.20	0.970	8.02	9.23	1.15	2
gs	winter	14,003	0.95	1.32	1.52	0.943	5.37	5.70	1.06	0
gs	spring	10,329	0.56	1.25	1.54	0.964	9.08	8.88	0.98	1
gs	summer	11,468	-0.40	1.50	2.01	0.930	12.43	12.60	1.01	3
gs	autumn	11,244	0.26	1.14	1.43	0.969	7.18	8.00	1.11	1
era5	winter	14,232	0.02	1.43	1.78	0.886	5.37	6.39	1.19	1
era5	spring	10,554	-0.24	1.61	2.08	0.924	9.08	9.21	1.01	2
era5	summer	12,283	-1.59	2.18	2.80	0.901	12.43	11.80	0.95	3
era5	autumn	12,112	-0.49	1.61	2.05	0.942	7.18	8.53	1.19	1

Table 5: Air-temperature agreement metrics.

5.2 Solar radiation

The radiation comparison will apply the Study 01 validity decisions before any alignment. It will therefore include corrected night-time zeros, retain valid daylight measurements, and exclude rejected samples and suspect days. Because radiation is strongly controlled by time of day and season, agreement will be examined both over the full valid set and within the seasonal daily cycle.

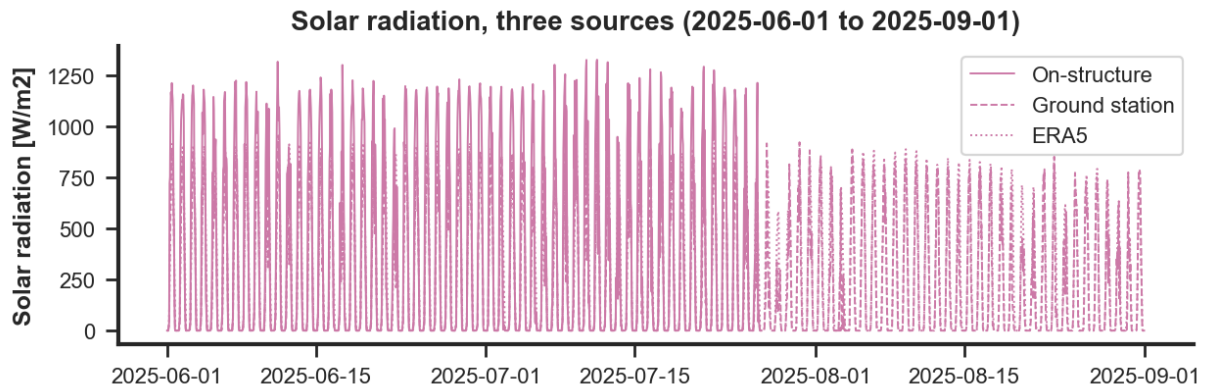


Figure 12: Aligned three-source solar-radiation time series over the valid on-structure timestamps.

Solar radiation over the certified window: each source against the wall

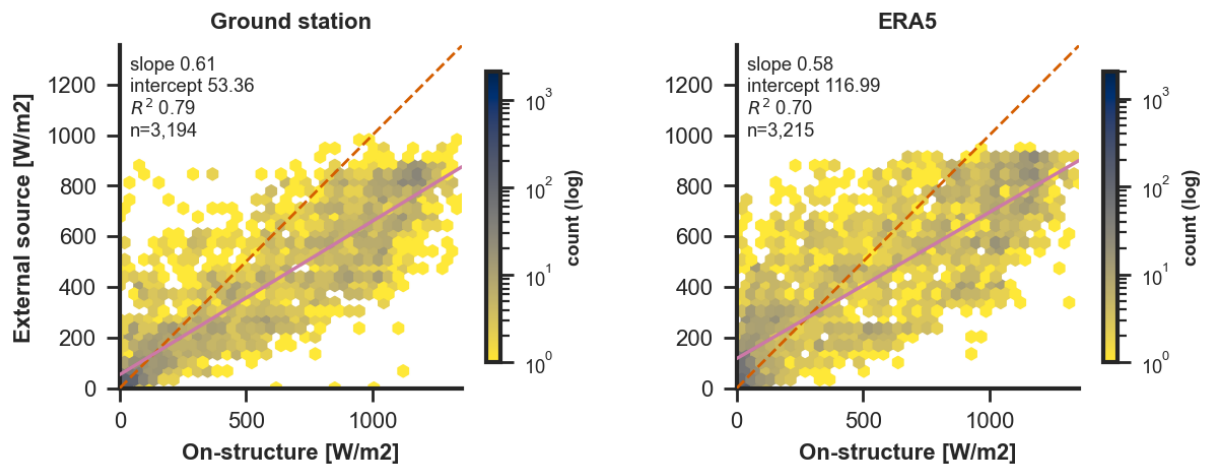


Figure 13: Direct comparison scatter plots of each solar-radiation proxy against the on-structure values.

Solar radiation, mean daily cycle by season and source

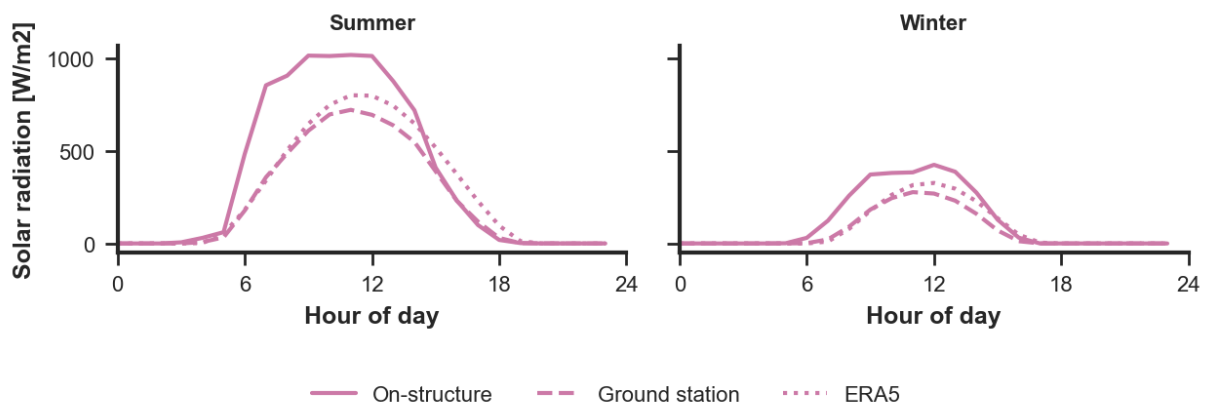


Figure 14: Solar-radiation diurnal cycles across sources.

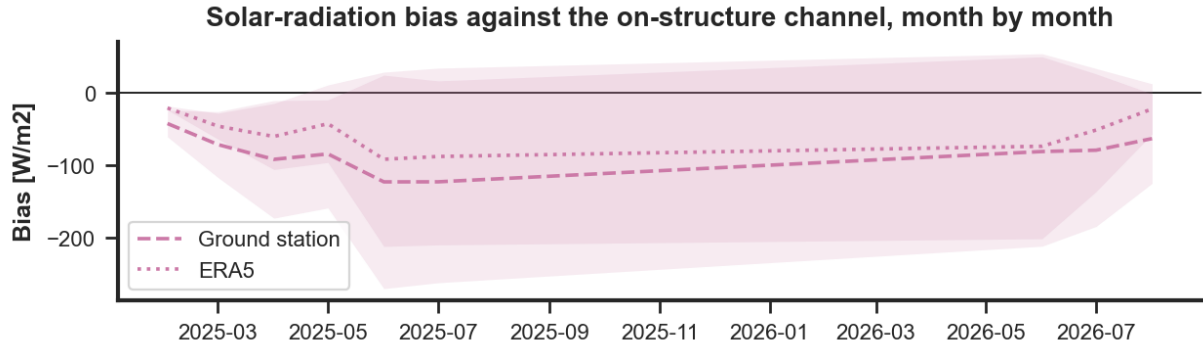


Figure 15: Solar-radiation stability and difference analysis.

Source	Season	N	Bias	MAE	RMSE	r	Amp (On)	Amp (Prx)	Amp Ratio	Phase D
gs	all	5,008	-87.7	117.7	204.0	0.925	1161.0	598.8	0.52	0
era5	all	5,043	-56.5	125.2	206.6	0.892	1161.0	609.7	0.53	0
gs	winter	195	-42.0	47.8	108.9	0.942	673.0	341.0	0.51	0
gs	spring	2,191	-82.2	96.5	176.4	0.941	1027.0	693.5	0.68	0
gs	summer	2,598	-98.3	140.3	228.5	0.917	1189.7	822.0	0.69	0
gs	autumn	24	185.1	185.4	299.0	0.918	31.9	500.0	15.67	0
era5	winter	195	-20.5	62.9	124.1	0.832	673.0	337.6	0.50	0
era5	spring	2,193	-49.2	106.7	185.6	0.888	1027.0	713.3	0.69	0
era5	summer	2,631	-67.6	144.5	225.7	0.893	1189.7	849.6	0.71	1
era5	autumn	24	201.4	202.2	328.9	0.949	31.9	490.3	15.36	0

Table 6: Solar-radiation agreement metrics.

5.3 Compatibility and substitution

The final assessment will state separately whether ERA5 or Meteosystem is a defensible substitute for on-structure air temperature and for on-structure solar radiation. A source may be useful without being interchangeable with the local measurement: a stable bias or amplitude difference may permit a documented transformation, whereas an unstable seasonal relation, inconsistent phase, or poor correspondence during the valid local window would rule out substitution. The verdict will specify the accepted source, any required transformation, the period over which the evidence applies, and the limitations carried into the later inclination analysis.

6 Conclusions

This section will summarise the character of the two external records and give the source verdict for each of the two forcings. It will distinguish evidence that supports filling the missing on-structure coverage from evidence that only supports using a source as a broader environmental covariate. Any unresolved metadata or unsupported substitution will remain explicit rather than being converted into an assumption.